

Schema Crush

AI-driven biomedical schema mapping to FHIR and bioinformatics graph components

Automating schema harmonization across clinical and research data ecosystems

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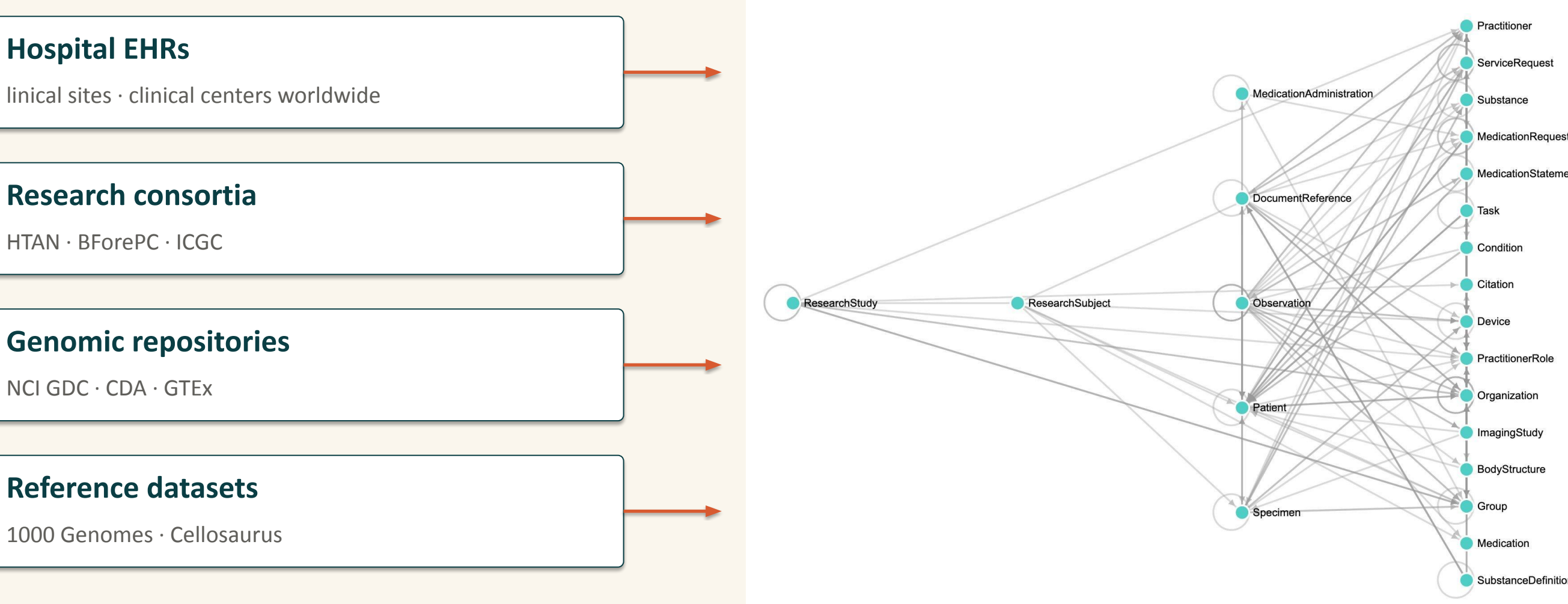


MOTIVATION & PROBLEM

Biomedical consortia produce massive, heterogeneous datasets using stand-alone schemas that block cross-institutional integration, EHR alignment, and federated AI/ML workflows.

Manual or semi-manual schema mapping takes **months per dataset**, spans **thousands of fields**, and yields **non-reproducible** results. Ex. the Human Tumor Atlas Network (HTAN)'s ongoing mappings toward a shared LinkML-based schema unify research data across NIH consortia, but extending these representations into FHIR enables interoperability with EHR systems and broader clinical data networks supporting scalable integration across both research and clinical domains.

Goal: Automate clinical + genomics schema mapping -> FHIR + genomics linked graph, with expert oversight.



heterogeneous sources -> FHIR R5 graph + linked genomics

APPROACH · CONTEXT-AWARE 3-TIER MAPPING

CSV Profiler is invoked by the CLI entry point in-process, or by a Claude/MCP client over stdio. The result sets the context across entity, field, and content tiers - a bioinformatician's first glance at data. Inside that context: 1253 expert-curated rules + embeddings + LLM, ensembled per tier. Confidence router -> accept / HITL / reject.

Entity tier
Ex: case -> Patient · BioBERT (semantic similarity) + expert-curated rules
Field tier
Ex: column alignment · Magneto (122 GDC patterns) + expert-curated rules
Content tier
Ex: coded values · SNOMED / LOINC + expert-curated rules

USE CASES

- Cross-institutional interoper**
faster onboarding of new sites with minimal integration effort.
- Federated ML enablement**
phenotype-consistent features across sites.
- Clinical ↔ research bridge**
EHR records into graph-ready research schemas.
- Cross-modal integration**
multi-omics alignment applications with clinical phenotypes.

ARCHITECTURE

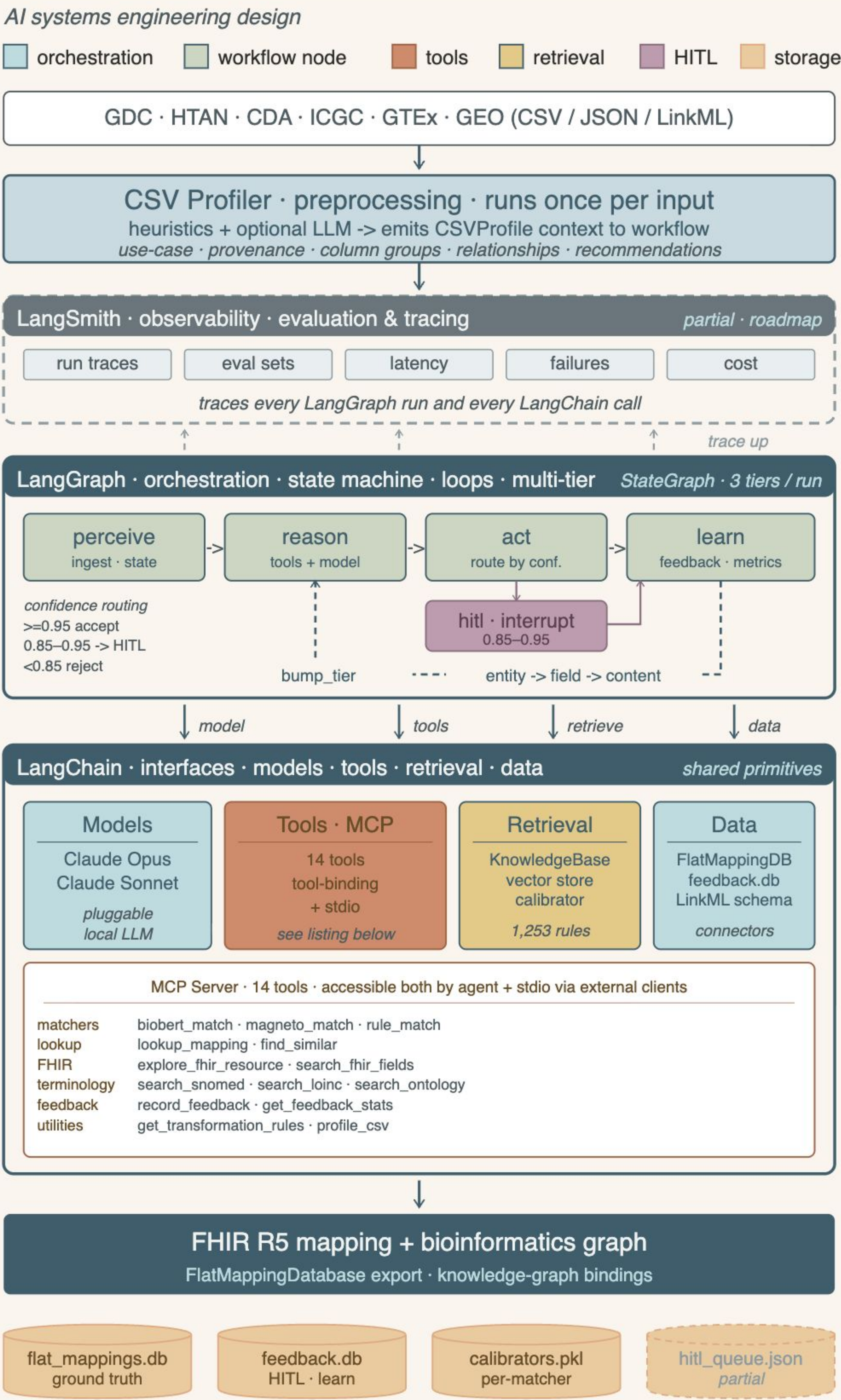


Figure 1: Technical architecture. A production, cost-efficient AI systems stack for schema harmonization producing graph-structured, ML/AI-ready FHIR output. A LangGraph StateGraph loops entity, field, and content tiers through five nodes (perceive, reason, act, HITL, learn). Three ensembled matchers (BioBERT, Magneto, RuleMatcher) score each candidate; LLM reasoning selects among them; a calibrated confidence router gates proposals to auto-accept, HITL review, or reject. Four persistence layers support the pipeline: curated mappings (SQLite), HITL feedback (SQLite), per-matcher calibrators (pickle), and ChromaDB. The same 14 tools are invoked both in-process (LangChain tool binding) and externally via MCP stdio, while typed FHIR references ingest directly into graph databases and ontology-coded features feed ML pipelines without per-site reconciliation.

PRELIMINARY RESULTS

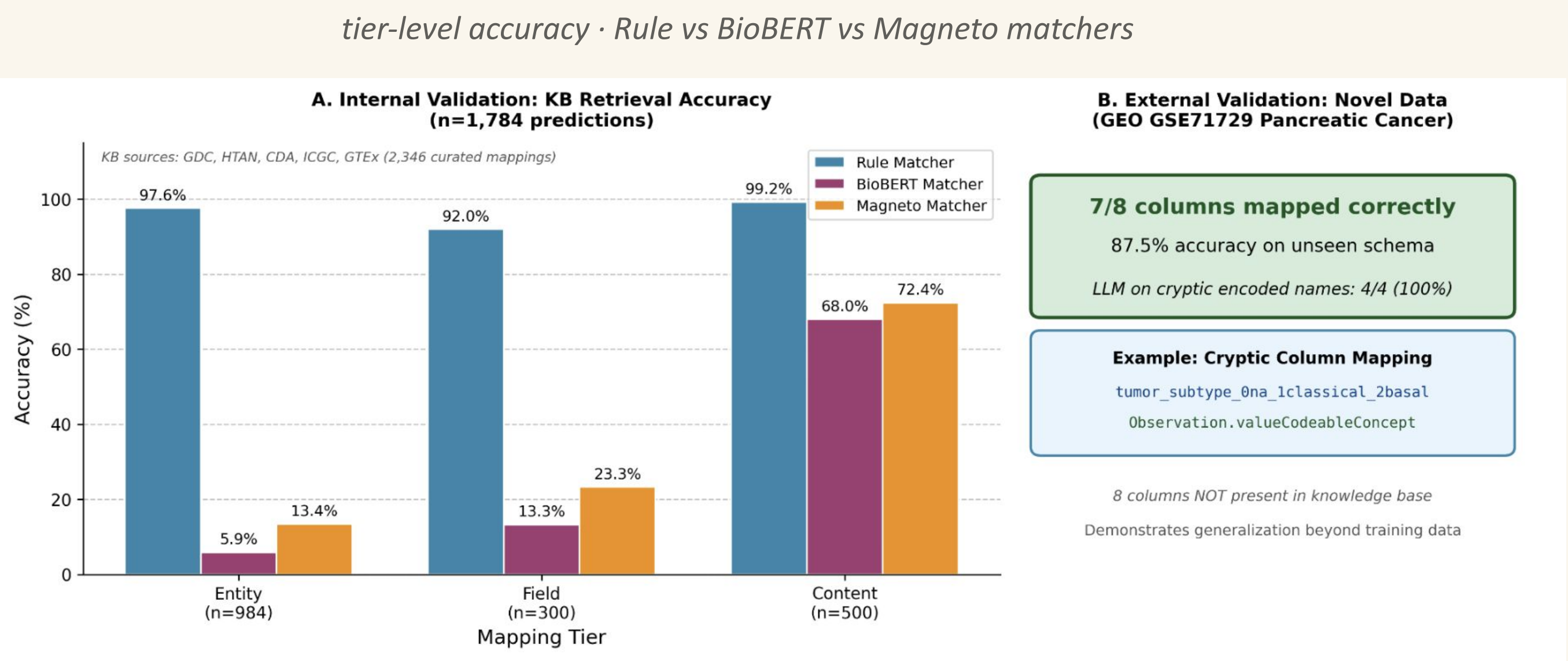


Figure 2: (A) Internal validation on 1,784 expert-curated mappings from five bioinformatics data repositories (GDC, HTAN, CDA, ICGC, GTEx). The Rule Matcher achieves 92-99% accuracy across entity, field, and content tiers by retrieving from a curated knowledge base. Embedding-based matchers (BioBERT, Magneto) complement rule-based matching for semantic similarity on novel terms. (B) External validation on GEO GSE71729 metadata demonstrates generalization to unseen schemas, achieving 87.5% accuracy on 8 columns absent from the knowledge base. LLM reasoning achieved 100% accuracy (4/4) on cryptic encoded column names demonstrating the system's ability to interpret non-standard nomenclature common in standalone research datasets.

EXAMPLE

1 · cryptic column · LLM resolves encoded names

input · GSE71729 clinical column
tumor_subtype_0na_1classical_2basal
↓ LLM-driven resolution
Observation.valueCodeableConcept

2 · Complex hierarchical structural schema mapping · AJCC staging · between schemas + FHIR

Condition ↔ Observation · parent-child grouping

Condition.stage[] contains:

- summary stage value (ex. Stage IIA)
- type SNOMED 1222593009 · tumor staging
- assessment[].ref → forward pointer to parent Observation

Parent Staging Observation:

- focus[].ref ← back-pointer to Condition
- hasMember[] → T · N · M · Grade child observations

DISCUSSION & FUTURE WORK

Schema Crush grounds each mapping in domain context; expert-curated rules, calibrated embeddings, and LLM reasoning act together to route only **uncertain cases to HITL**.

Schema Crush transforms schema harmonization from a manual bottleneck into a **scalable, cost-efficient step**, producing **graph-structured data** for downstream ML and AI **across clinical and genomics domains**. Each curator decision (accept/edit/reject) becomes training signal, feeding a continuously learning, explainable system through persistent feedback and adaptive weighting.

This turns schema mapping into a reusable, continuously improving system for data integration.

Next: Durable orchestration + tracing + learning layer (checkpointed HITL, per-mapping evals in prod), ontology expansion, few-shot retrieval (ChromaDB), and cell level integration for **cellular-resolution multimodal integration + analytics**.

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Planned release is to PyPI and the public MCP server registry.

HTAN · BForePC · GA4GH Working Group